COMMONWEALTH OF AUSTRALIA

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Southern Cross University

First Part/Chapter of the Autonomic Nervous System (ANS) Pharmacology

Introductory Pharmacology
PHAR2001

Dr Ken Wojcikowski



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Autonomic Nervous System Pharmacology

Chapter Approach for the ANS:

1. Introduction
2. Adrenergic pharmacology (Agonists)
3. Adrenergic pharmacology (Antagonists)
4. Cholinergic pharmacology



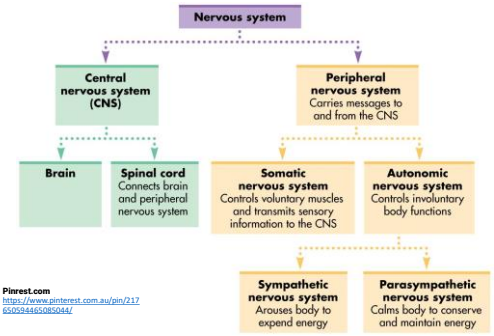
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Foundational Information for Understanding Autonomic Nervous System (ANS) Pharmacology

Lesson 1:

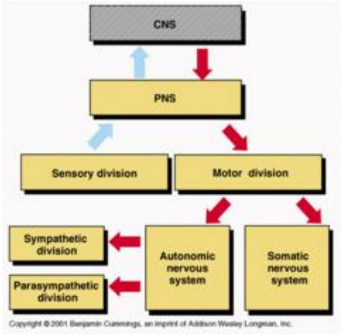
Overview of the autonomic nervous system

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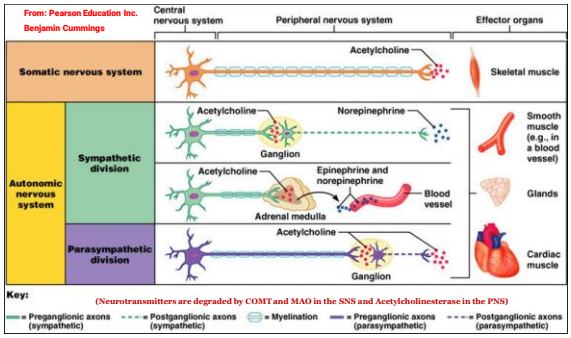


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https://www.pinterest.com.au/pin/217650934465085044/

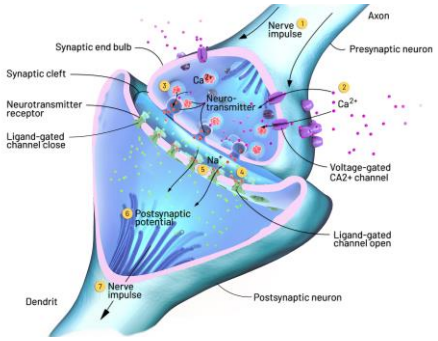
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Foundational Information for Understanding ANS Pharmacology

Lesson 2:

Using logic to understand the divisions of the ANS

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The sympathetic system

'Fight or flight' responses

- Rapid heart rate
- No need to digest or urinate
- Need to dilate pupils

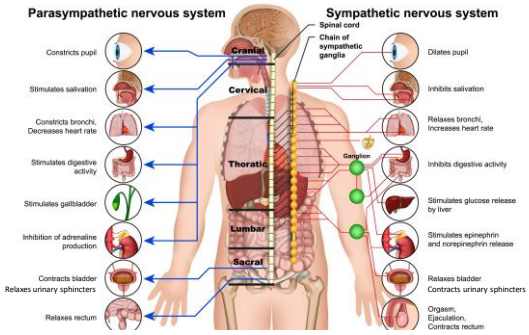
Think about the last time a bear chased you…

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https://www.telegraph.co.uk/news/newstopics/howaboutthat

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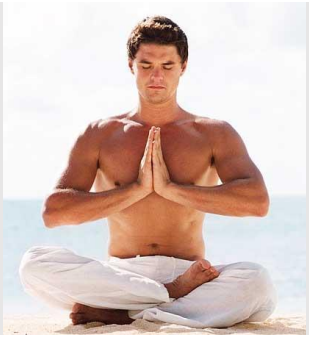
The parasympathetic system

The *'rest and digest'* system

- Heart rate slows
- Digestion begins
- Pupils constrict

Think about the last time you were able to chill out (*well, try to remember*)

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Foundational Information for Understanding ANS Pharmacology

Lesson 3:

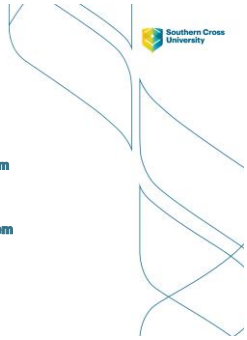
Overview of the drugs of the ANS

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ANS pharmacology has one simple rule

Drugs that affect the autonomic nervous system have one of four actions

1. They can increase the sympathetic system response
2. They can increase the parasympathetic system response
3. They can decrease the sympathetic system response
4. They can decrease the parasympathetic system response



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In medicine, there are always exceptions to the rule...

Drugs that affect the autonomic nervous system have one of four actions

1. They can increase the sympathetic system response
2. They can increase the parasympathetic system response
3. They can decrease the sympathetic system response
4. They can decrease the parasympathetic system response

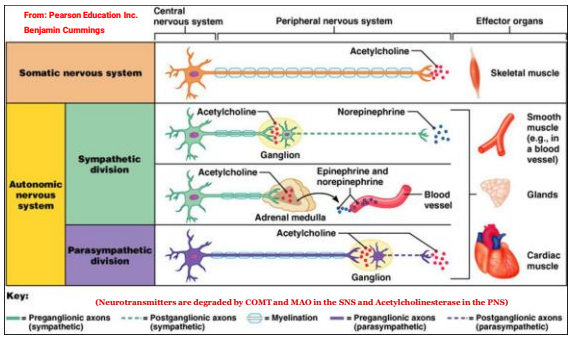
EXCEPT:

- Alpha 2 receptors (sympathetic system)
- Central nicotinic receptors (parasympathetic system)

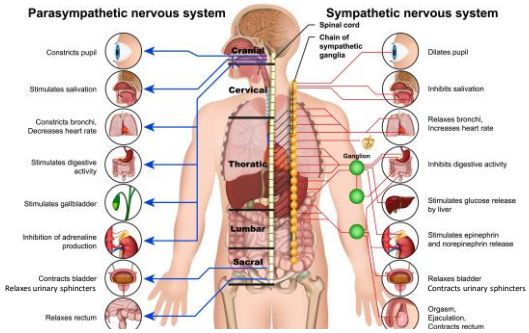
LET'S IGNORE THE EXCEPTIONS UNTIL LATER
(postpone readings until the appropriate time)



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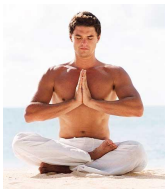
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Functions of ANS	
Sympathetic	Parasympathetic
▸ Heart	▸ Heart
▸ ↑ heart rate	▸ ↓ heart rate
▸ ↑ force of contraction	▸ ↓force of contraction
▸ Blood vessels	▸ Blood vessels
▸ *Constriction*	▸ No effect
▸ Lungs	▸ Lungs
▸ Bronchodilation	▸ Bronchoconstriction
▸ GIT	▸ GIT
▸ ↓ motility	▸ ↑ motility
▸ Sphincter contraction	▸ Sphincter relaxation
▸ Decreased secretions	▸ Increased secretions

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Autonomic Nervous System (ANS)
Sympathetic vs Parasympathetic





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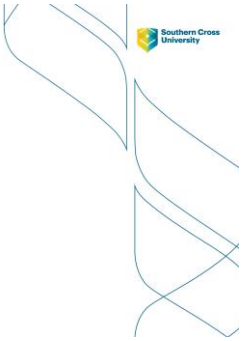
Foundational Information for Understanding ANS Pharmacology
Lesson 4:
The selective nature of ANS receptors

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'Selective' SNS receptors: Alpha 1 receptors

Location of receptors:

- Blood vessels
- Smooth muscles of the GIT
- Sweat glands
- Radial muscles of the eye
- Sphincters of the bladder

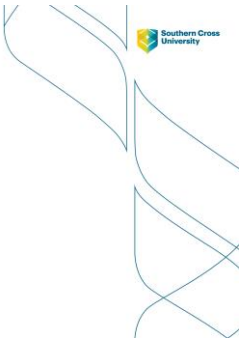


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'Selective' SNS receptors: Beta 1 receptors

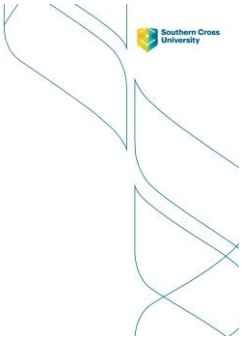
Location of receptors:

- Heart
- Git
- Kidneys



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'Selective' SNS receptors: Beta 2 receptors



Location of receptors:

- Lung
- Mast cells
- Uterus
- Skeletal muscle

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References

- Goldsy RA et Al. Immunology 5th ed. WH Freeman Publishers, 2003
- Marieb E. Human Anatomy and Physiology, 5th ed. Benjamin Cummings, San Francisco, 2001
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- Martini FH, et al. Visual Anatomy and Physiology 1st ed. Benjamin Cummings, 2011
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- The PowerPoint and associated Video were edited and produced by Dr. Ken Wojcikowski

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